

To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture.

```
priyanka@DESKTOP-HOISO7A:~$ uname
Linux
priyanka@DESKTOP-HOISO7A:~$ uname -a
Linux DESKTOP-HOISO7A 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
priyanka@DESKTOP-HOISO7A:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

priyanka@DESKTOP-HOISO7A:~$ git --version
git version 2.53.0
priyanka@DESKTOP-HOISO7A:~$ mkdir -p ~/OS_Skill/Week-1
cd ~/OS_Skill/Week-1
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ pwd
/home/priyanka/OS_Skill/Week-1
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ mkdir -p src include bin
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ ls
bin include src
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ Week-1
Week-1: command not found
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ pwd
/home/priyanka/OS_Skill/Week-1
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
```

Initialized empty Git repository in /home/priyanka/OS_Skill/Week-1/.git/

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ git status
On branch master
```

No commits yet

nothing to commit (create/copy files and use "git add" to track)

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ cd src
```

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ nano process.c
```

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ ls
```

process.c

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ gcc process.c -o ../bin/process
```

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ ls ../bin
```

process

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ ../bin/process
```

Hello from Operating System Lab!

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ ../bin/process
```

Hello from Operating System Lab!

Code:

```
priyanka@DESKTOP-HOISO7A: ~$ nano process.c
GNU nano 8.7.1 process.c *
#include <stdio.h>

int main()
{
    printf("Hello from Operating System Lab!\n");

    return 0;
}
```

Build Initial Makefile

Code:

```
GNU nano 8.7.1 Makefile *
CC = gcc

all:
    $(CC) src/process.c -o bin/process

clean:
    rm -f bin/process
```

Output:

```
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1/src$ cd ..
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ nano Makefile

priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ make
gcc src/process.c -o bin/process
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ ls bin
process
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ ./bin/process
Hello from Operating System Lab!
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ make clean
rm -f bin/process
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ ls bin
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ make
gcc src/process.c -o bin/process
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ make
gcc src/process.c -o bin/process
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-1$ |
```